Evaluation Discipline for Synthetic Attrition Classification: Metric Bounding in Small-Sample Tabular ML

Shaurya Gupta

Independent Researcher

Email: shauryagupta042@gmail.com

Abstract

Small organizational ML benchmarks can support useful evaluation work, but their outputs are easily inflated when discrimination scores, explanation plots, clustering metrics, and subgroup statistics are read beyond the evidence that generated them. This article develops a disciplined evaluation logic for constrained tabular benchmarks and demonstrates it on the IBM HR Analytics Employee Attrition dataset, a synthetic pedagogical file with 1,470 observations and 35 variables (IBM Data Scientists, 2015). The pipeline combines leakage-aware preprocessing, arithmetic feature derivation, clustering diagnostics, supervised benchmarks, calibration, permutation importance, SHAP inspection, subgroup metric computation, and seed-sensitivity checks. CatBoost achieved ROC-AUC = 0.8181 on the held-out split, with bootstrap 95% interval [0.7394, 0.9012]. DBSCAN returned silhouette = 0.2729 with 17.3% noise, providing little support for stable segmentation in the engineered feature space. The gender prediction-association diagnostic returned Cramer's V = 0.0001; within this benchmark, that statistic is a metric-execution result rather than a sociotechnical fairness finding. The contribution is a provenance-aware reporting template for small-sample organizational ML: evaluation outputs remain informative, but their interpretation is bounded by dataset realism, model uncertainty, metric design, and the absence of causal or deployment evidence.

Keywords: tabular classification; synthetic benchmark; attrition prediction; calibration; CatBoost; SHAP; clustering diagnostics; fairness metrics; MLOps evaluation

1 Introduction

Applied HR-AI evaluation often moves faster than its evidence base. A classifier produces a plausible ROC-AUC; a feature-attribution plot supplies a ranking; a clustering algorithm returns groups; subgroup tables appear to add fairness awareness. Each output can be technically correct and still exceed what the data can support. The methodological problem is not the existence of small benchmarks. It is the conversion of benchmark artifacts into organizational claims without adequate constraints on interpretation.

The IBM HR Analytics Employee Attrition dataset is useful precisely because its limits are visible. It contains 1,470 rows, 35 columns, a binary attrition label, and no observed employment histories, intervention records, grievance processes, labor-market shocks, or documented managerial decisions (IBM Data Scientists, 2015). Used as direct evidence about workplaces, it would be too thin. Used as a constrained evaluation surface, it can clarify how predictive performance, attribution, clustering, calibration, subgroup association, and robustness evidence should be bounded.

The study addresses a reporting problem that recurs across applied machine learning: benchmark outputs acquire more authority than their provenance warrants. In organizational domains, that slippage is especially consequential because the target labels appear socially meaningful even when the file is synthetic or pedagogical. Evaluation design must therefore encode interpretive boundaries before results are narrated as managerial knowledge.

The contribution is an evaluation discipline for small-sample synthetic organizational ML benchmarks. It does not rest on algorithmic novelty, causal identification, or deployment validation. It rests on a narrower but necessary claim: modest benchmarks remain analytically useful when the workflow reports uncertainty, separates model behavior from causal explanation, treats subgroup metrics as context-dependent, and refuses to turn weak geometric structure into employee categories.

2 Related Work

Attrition prediction has often been framed as a supervised learning problem. Logistic regression, random forests, gradient boosting, and related estimators can separate observed labels when the feature matrix contains predictive signal (Breiman, 2001; Friedman, 2001; Prokhorenkova et al., 2018). Discrimination performance is informative, but it is incomplete when reported without uncertainty, calibration, provenance, and limits on downstream interpretation.

HR analytics research has warned that workplace data is not neutral administrative residue. Compensation, tenure, department, promotion history, work allocation, and evaluation scores may encode power, hierarchy, and prior managerial decisions (Angrave et al., 2016; Marler and Boudreau, 2017; Tambe et al., 2019). Synthetic benchmarks occupy a different evidentiary category. They can test pipeline behavior, but they cannot be read as organizational records unless their generator models the relevant social and institutional processes.

Explainability methods expose fitted-model behavior. SHAP and local explanation techniques can show how features contribute to predictions inside a trained model (Lundberg and Lee, 2017; Ribeiro et al., 2016). Their interpretive status depends on the dataset. If `MonthlyIncome` dominates a synthetic attrition classifier, the attribution result identifies a strong numerical dependency in the fitted benchmark system; it does not, by itself, identify an intervention target.

Fairness research makes a stronger demand than subgroup tabulation. Protected-class metrics matter when the data-generating process reflects allocation systems, social histories, proxy variables, institutional feedback, and contested harms (Barocas and Selbst, 2016; Hardt et al., 2016; Mehrabi et al., 2021). On a synthetic file with undocumented bias generation, a Cramer's V statistic is best interpreted as a subgroup reporting exercise. Its value lies in making the distinction between metric computation and fairness evidence explicit.

Clustering has parallel interpretive risks. Internal validity scores summarize geometry in a feature representation; they do not validate psychological categories. A density-based algorithm such as DBSCAN may detect non-convex local density structure, while silhouette scoring rewards compact separation around centroids. A provenance-aware evaluation therefore reports the mismatch openly and resists the common slide from cluster labels to employee typologies.

3 Materials and Methods

3.1 Data

The analysis uses the IBM HR Analytics Employee Attrition dataset (IBM Data Scientists, 2015). The file contains 1,470 complete rows and 35 variables. The target is `Attrition`. The dataset is synthetic and pedagogical, making it suitable for bounded evaluation rather than workplace inference.

The empirical design follows from that premise. Results are interpreted as properties of the benchmark under a specified pipeline and are not generalized to firms, departments, occupations, industries, or labor markets.

3.2 Preprocessing

Identifier and constant fields are removed before modeling. Numeric variables are imputed and standardized. Categorical variables are imputed and one-hot encoded. The train/test split is stratified by the attrition label. Transformations are fit on the training partition before application to held-out data.

Direct protected or sensitive attributes are retained for subgroup metric computation, with interpretation restricted to the benchmark context.

3.3 Arithmetic feature derivation

Eight derived variables are computed from existing columns: Overtime Frequency Index from `OverTime`; Satisfaction-Job Level Index from `JobSatisfaction` and `JobLevel`; Tenure-to-Promotion Ratio from `YearsAtCompany` and `YearsSinceLastPromotion`; Overtime-Balance Ratio from `OverTime` and `WorkLifeBalance`; Compensation Progression Index from `MonthlyIncome` and `PercentSalaryHike`; Performance-Compensation Alignment Index from `PerformanceRating` and `PercentSalaryHike`; Work Arrangement Index from `BusinessTravel` and `WorkLifeBalance`; and Multi-Satisfaction Composite from ordinal satisfaction variables.

These variables are arithmetic transformations rather than latent constructs or psychometric scales.

3.4 Clustering

KMeans, Gaussian mixture models, agglomerative clustering, and DBSCAN are evaluated. Metrics include silhouette score, Davies–Bouldin index, Calinski–Harabasz score, cluster profiles, demographic association checks, and seed sensitivity.

The selected DBSCAN run uses eps = 2.0 and min_samples = 10. The parameterization is reported as an empirical configuration, not as an optimized density model derived from a formal topological analysis.

3.5 Supervised classification

Models include a majority baseline, logistic regression, random forest, and CatBoost gradient boosting (Breiman, 2001; Friedman, 2001; Prokhorenkova et al., 2018). Both salary-only and full-feature specifications are estimated. Metrics include ROC-AUC, average precision, Brier score, accuracy, F1, calibration curves, bootstrap intervals, and seed-sensitivity checks.

Synthetic oversampling is not used. This choice keeps the class distribution in the fitted probability outputs closer to the observed benchmark split. It is not a claim that oversampling is methodologically invalid in general.

3.6 Attribution and subgroup metrics

Permutation importance and SHAP are used to inspect fitted-model behavior (Lundberg and Lee, 2017; Ribeiro et al., 2016). The outputs report how the model uses columns in the benchmark feature space.

Subgroup metrics are computed across gender, age band, department, compensation band, and marital status. These computations demonstrate subgroup reporting under constrained data provenance.

4 Clustering Results

DBSCAN identified two non-noise groups and assigned 17.3% of observations to noise. Among non-noise observations, silhouette was 0.2729, Davies–Bouldin was 1.4146, and Calinski–Harabasz was 341.59. Table 1 reports the selected metrics.

The result is weak. A silhouette value of 0.2729 does not support separable convex manifolds in the engineered feature space. Because DBSCAN is density-based and silhouette scoring is less natural for irregular density clusters, the score should be interpreted with caution. The reported geometry supports profile inspection only, not employee segmentation.

Table 1. Selected clustering metrics for the reported DBSCAN configuration.

Algorithm	Clusters	Noise fraction	Silhouette	Davies–Bouldin
DBSCAN	2	0.1735	0.2729	1.4146

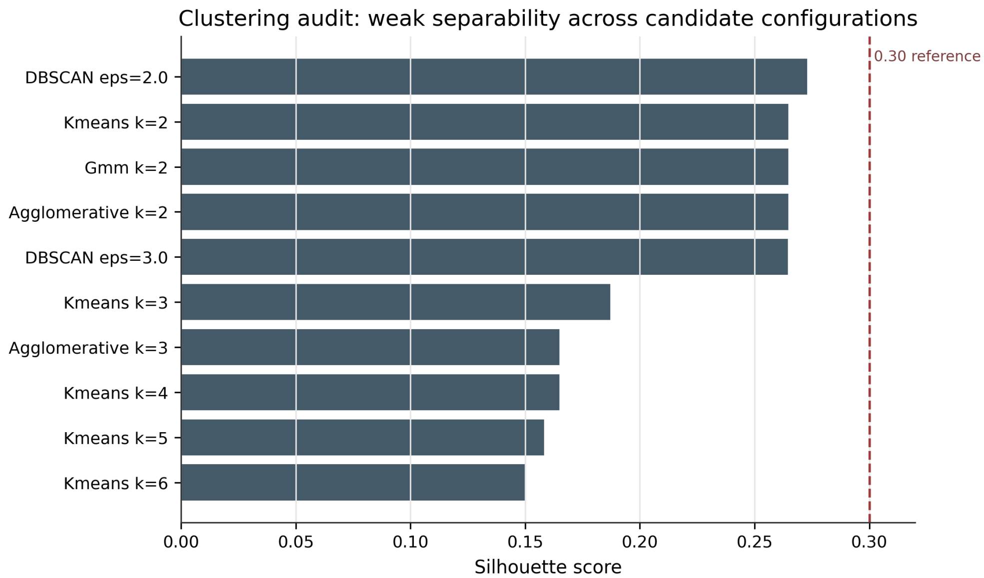

Figure 1. Clustering score comparison. The reported DBSCAN run provides limited evidence of separable structure in the engineered representation.

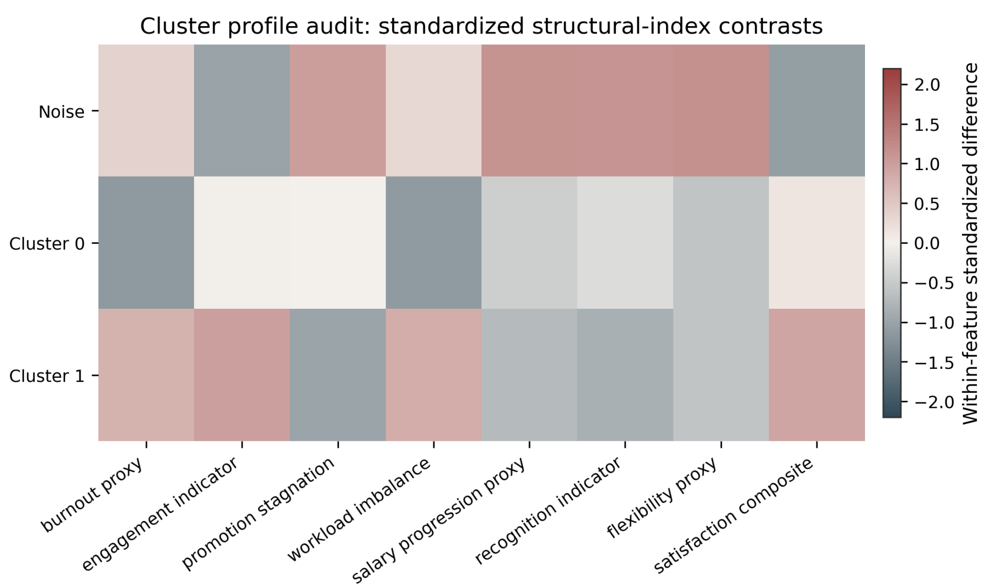

Figure 2. Cluster profile heatmap. The profiles summarize arithmetic feature differences under the reported clustering configuration.

5 Predictive Results

The full-feature CatBoost model achieved ROC-AUC = 0.8181. The salary-only CatBoost model achieved ROC-AUC = 0.6688. Logistic regression and random forest produced ROC-AUC = 0.7914 and ROC-AUC = 0.7902, respectively. Table 2 reports selected results.

The bootstrap 95% interval for the full-feature CatBoost ROC-AUC was [0.7394, 0.9012]. This interval quantifies variation under resampling of the benchmark evaluation set and provides a useful guard against overreading a single split.

The full-feature CatBoost model also achieved average precision = 0.5626, Brier score = 0.0996, accuracy = 0.8639, and F1 = 0.3939 at the default threshold. The F1 value is modest despite high accuracy, which is expected under class imbalance.

Table 2. Selected predictive benchmark results.

Feature set	Model	ROC-AUC	Average precision	Brier score	Accuracy
All features	CatBoost	0.8181	0.5626	0.0996	0.8639
Salary only	CatBoost	0.6688	0.2750	0.1338	0.8265
All features	Logistic regression	0.7914	0.5483	0.1594	0.7789
All features	Random forest	0.7902	0.5062	0.1118	0.8605

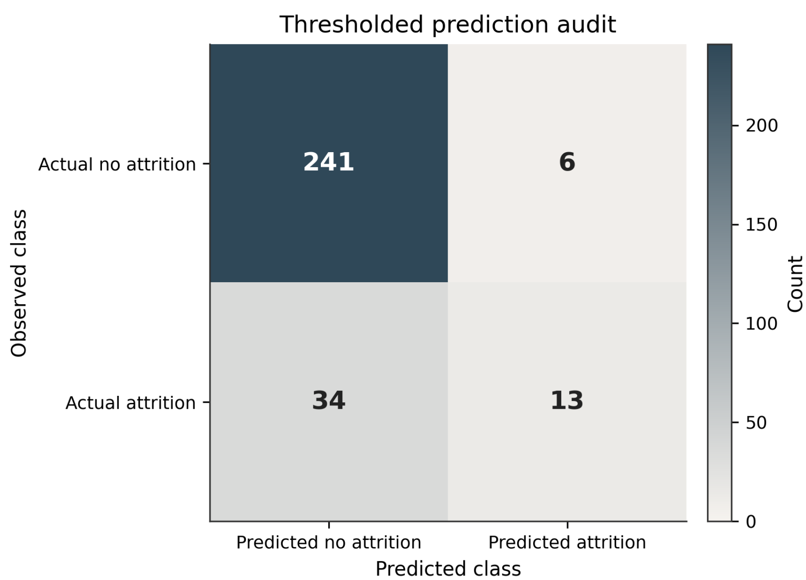

Figure 3. Confusion matrix for the selected classifier at the default threshold.

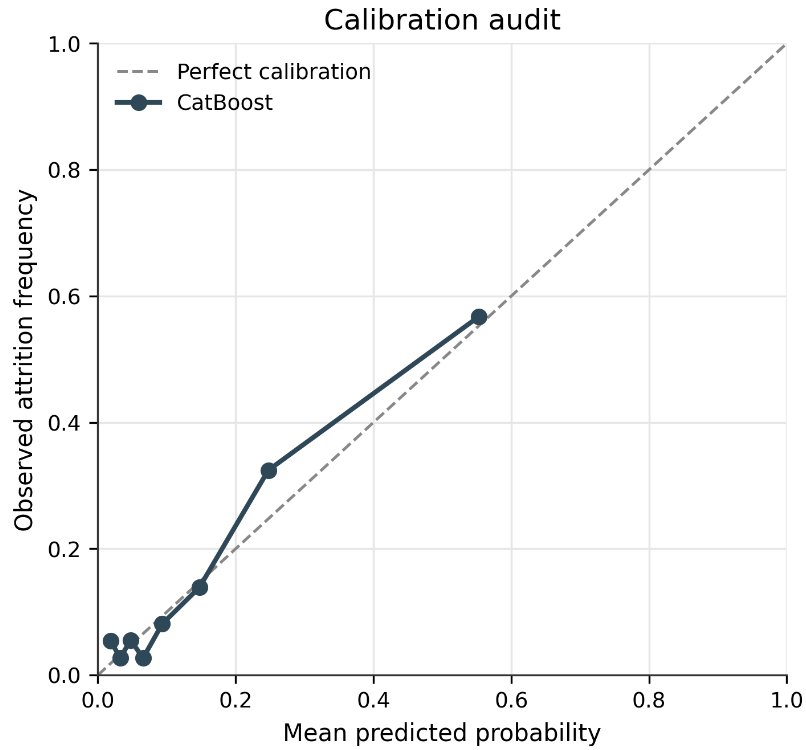

Figure 4. Calibration curve for the selected classifier under the benchmark split.

6 Attribution and Subgroup Metrics

Permutation importance and SHAP identify the features most used by the fitted classifier. Compensation-related variables are prominent. In this benchmark, that result is best read as model inspection: the fitted estimator assigns substantial predictive weight to compensation-linked columns, while causal interpretation remains outside the design.

The gender prediction-association diagnostic returned Cramer's V = 0.0001. Under this file and fitted model, the recorded gender column has negligible association with the computed prediction grouping. Since the benchmark does not document historical bias, organizational exclusion, managerial allocation, or proxy discrimination, the statistic should be read as subgroup reporting rather than a fairness conclusion.

The compensation band association was larger, with Cramer's V = 0.2847. In observed HR data, such a pattern would require domain review because compensation, seniority, role, tenure, and protected status can be entangled.

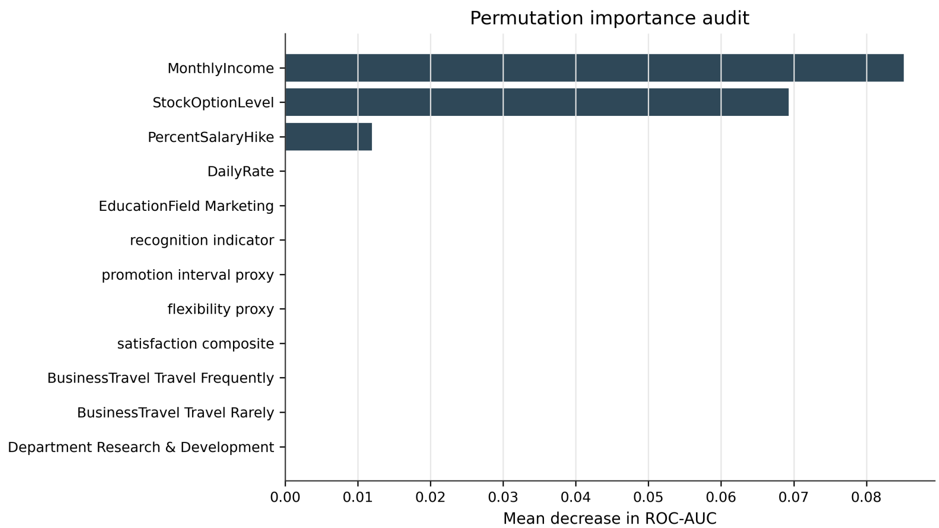

Figure 5. Permutation importance for the selected model. The figure reports predictive sensitivity, not causal effect.

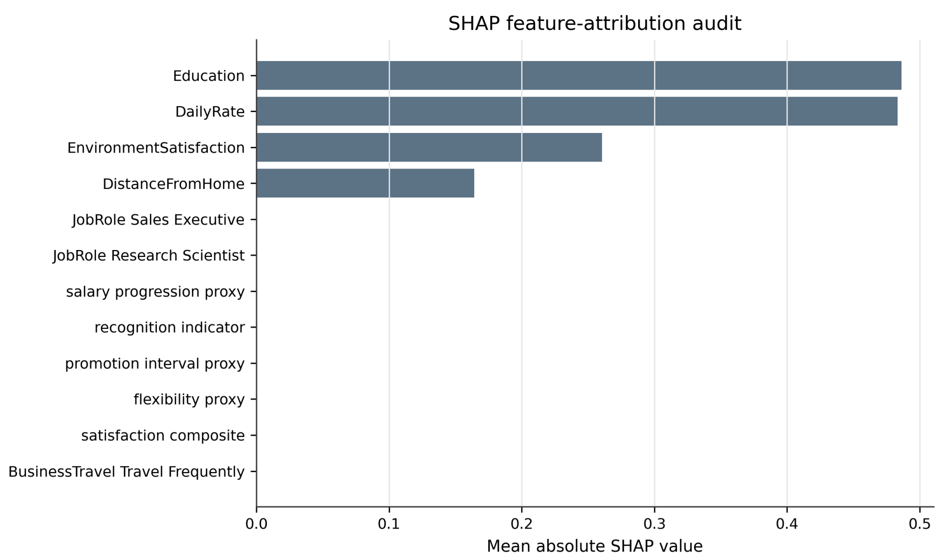

Figure 6. SHAP summary for the selected model. The values describe fitted-model attribution in the benchmark feature space.

Table 3. Selected subgroup metric outputs.

Metric type	Attribute	Disparate-impact ratio	Cramer's V
Predicted-positive rate	Gender	0.8961	–
Prediction association	Gender	–	0.0001
Prediction association	Age band	–	0.1415
Prediction association	Compensation band	–	0.2847
Predicted-positive rate	Marital status	0.2431	–

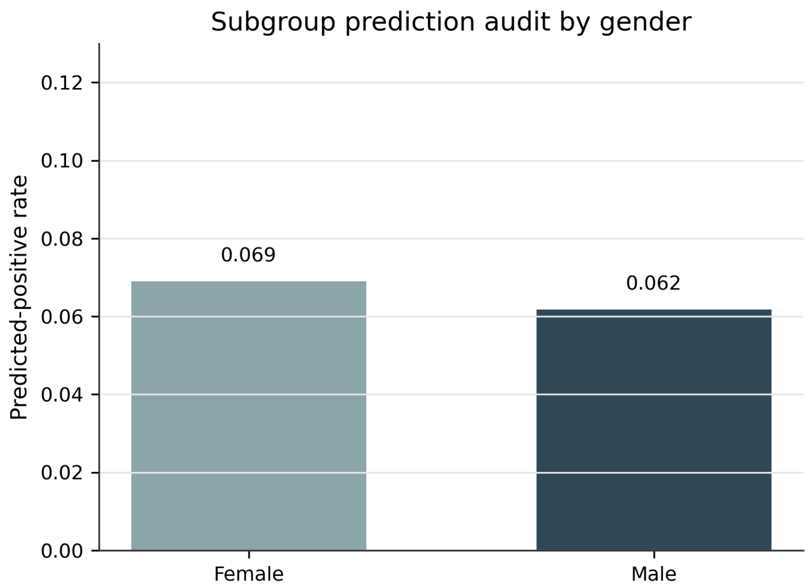

Figure 7. Gender-stratified predicted-positive rates under the benchmark prediction rule.

7 Seed Sensitivity

Repeated splits expose variation in reported performance. Figure 8 shows ROC-AUC variation across random seeds for logistic regression. The result supports uncertainty-aware reporting for small tabular benchmarks.

8 Discussion

The analysis demonstrates how a small organizational ML benchmark can be useful without being overextended. The value is not the CatBoost score alone, the SHAP display, the cluster labels, or the subgroup table. It is the evaluation logic that holds these outputs in relation to one another: performance is reported with uncertainty, attribution is separated from causality, clustering is judged against weak geometric evidence, and subgroup metrics are interpreted through data provenance.

CatBoost separates the attrition label better than the salary-only benchmark, with ROC-AUC = 0.8181 versus 0.6688. That result is a valid benchmark finding. Its interpretation is bounded by the file's synthetic provenance, the small positive class, the lack of external validation, and the absence of a documented generator in the analysis. The point is not to diminish the score. The point is to prevent the score from doing work it was not designed to do.

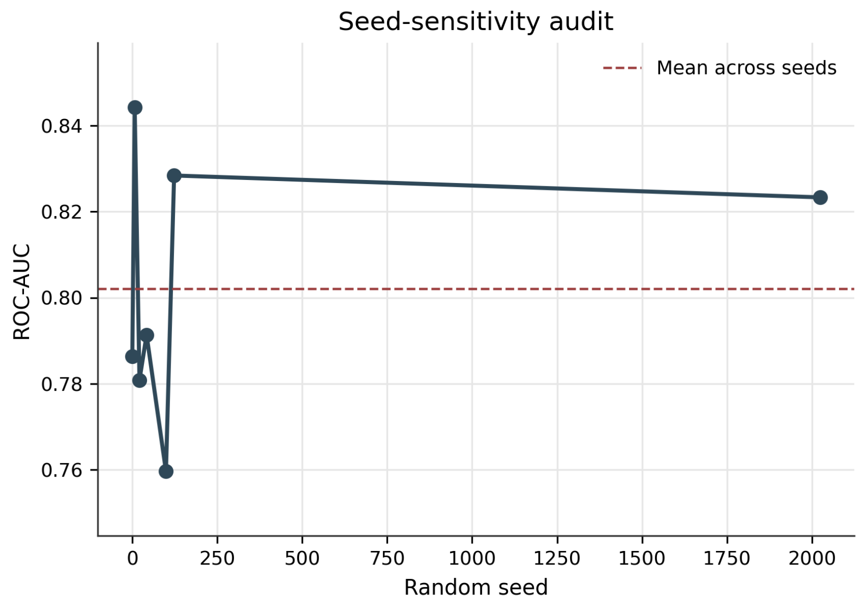

Figure 8. ROC-AUC variation across random seeds for logistic regression.

The clustering result sharpens the same logic. The engineered variables do not yield a robust segmented representation under the reported DBSCAN configuration. The low silhouette score, the noise fraction, and the algorithm-metric mismatch make employee typologies inappropriate. That is still an empirical result: the tested representation supports classification better than unsupervised segmentation.

The attribution and subgroup analyses extend the reporting discipline. Compensation-linked variables drive fitted-model behavior, which is relevant for understanding prediction mechanics. Cramer's V and predicted-positive-rate tables make subgroup patterns visible under the benchmark prediction rule. These outputs become stronger, not weaker, when their evidentiary status is stated precisely.

The broader implication concerns evaluation practice in applied AI. Synthetic and pedagogical datasets are often treated as either harmless demonstrations or invalid artifacts. A more useful position is available. They can support methodological learning when the evaluation process encodes provenance realism, uncertainty visibility, metric-context separation, and disciplined limits on inference. Under that standard, small benchmarks become sites for training interpretation, not vehicles for inflated organizational claims.

For HR analytics, this matters because the vocabulary of prediction can quickly acquire managerial force. A probability score resembles risk knowledge; a cluster label resembles a workforce segment; a subgroup table resembles a fairness audit. Evaluation discipline interrupts that conversion. It asks what the dataset can support, what the metric actually measures, and what additional evidence would be required before any operational interpretation could follow.

9 Limitations

The dataset is synthetic. This is the main scope condition.

The generator is not analyzed. If the attrition label was created from deterministic or semi-deterministic relationships among compensation, overtime, satisfaction, and tenure variables, the model may be recovering those rules rather than learning any pattern with external meaning.

The sample is small and imbalanced. A single split can produce unstable performance estimates.

The feature derivations are arithmetic. They should not be described as behavioral constructs, structural measures, or psychometric indices.

The clustering exercise does not establish separable structure. The silhouette score is low, and the use of silhouette for DBSCAN has known interpretive friction.

The subgroup metrics are benchmark-context outputs. Fairness analysis in observed HR systems would require stronger provenance, stakeholder-defined harms, proxy analysis, and mitigation testing.

The analysis is not causal. It cannot estimate why attrition occurs, whether compensation changes would alter attrition, or whether any intervention would retain workers.

10 Ethical and Reporting Position

No output from this pipeline should be used for employment decisions. The analysis is retrospective, benchmark-bound, and not externally validated.

The appropriate use is methodological. It shows how analysts can bound model claims, inspect attribution, compute subgroup metrics, and report uncertainty without converting benchmark scores into workplace conclusions.

Any real attrition system would require observed longitudinal data, documented provenance, intervention records, legal review, stakeholder input, proxy analysis, subgroup monitoring, model-card documentation, appeal channels, and post-deployment harm tracking (Mitchell et al., 2019; Raji et al., 2020).

11 Conclusion

The IBM attrition benchmark supports a compact but meaningful evaluation exercise when its outputs are kept within their evidentiary frame. A CatBoost classifier achieved ROC-AUC = 0.8181, with bootstrap 95% interval [0.7394, 0.9012]. DBSCAN produced silhouette = 0.2729 and 17.3% noise. The gender prediction-association diagnostic returned Cramer's V = 0.0001. Read together, these results show why applied AI evaluation should report more than predictive discrimination.

The supervised model identifies separable signal in the benchmark. The clustering result limits segmentation claims. The attribution outputs describe fitted-model behavior. The subgroup metrics demonstrate how prediction patterns can be stratified while remaining dependent on data provenance. None of these outputs needs to be inflated to be useful.

The contribution is a disciplined evaluation logic for synthetic, imbalanced, small-sample organizational ML benchmarks. Such benchmarks are not worthless; they are fragile evidence substrates. Their value depends on reporting practices that make uncertainty visible, preserve the distinction between model behavior and causal explanation, and prevent metric outputs from acquiring unwarranted organizational authority. In applied AI, methodological credibility begins when evaluation does not merely generate results, but governs what those results are allowed to mean.

Disclosure of Interest

The author reports no competing interests.

AI Assistance Disclosure

Generative AI tools were used for workflow orchestration, formatting assistance, code-development support, and language editing. The author reviewed the empirical outputs, methodological choices, and final manuscript.

Data Availability Statement

The IBM HR Analytics Employee Attrition dataset used in this study is publicly available through Kaggle: https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset

Supplementary materials, manuscript artifacts, and supporting documentation are available in the associated public repository: https://github.com/SilverCoin256/employee-incentive-ai-research

CRediT Author Statement

Shaurya Gupta: Conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing–original draft, writing–review and editing, visualization, and project administration.

Supplementary Appendix Reference

The supplementary appendix documents empirical outputs, arithmetic feature definitions, clustering results, and subgroup metrics. It is provided as `supplementary_appendix.md` in the submission package.

References

IBM Data Scientists. (2015). *IBM HR Analytics Employee Attrition & Performance* [Synthetic pedagogical dataset]. Kaggle. https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset

Angrave, D., Charlwood, A., Kirkpatrick, I., Lawrence, M., & Stuart, M. (2016). HR and analytics: Why HR is set to fail the big data challenge. *Human Resource Management Journal*, 26(1), 1–11. doi:10.1111/1748-8583.12090

Barocas, S., & Selbst, A. D. (2016). Big data's disparate impact. *California Law Review*, 104(3), 671–732. doi:10.2139/ssrn.2477899

Breiman, L. (2001). Random forests. *Machine Learning*, 45, 5–32. doi:10.1023/A:1010933404324

Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., & Gulin, A. (2018). CatBoost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems*, 31. https://proceedings.neurips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5), 1189–1232. doi:10.1214/aos/1013203451

Hardt, M., Price, E., & Srebro, N. (2016). Equality of opportunity in supervised learning. In *Proceedings of the 30th International Conference on Neural Information Processing Systems* (pp. 3323–3331). doi:10.5555/3157382.3157469

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Proceedings of the 31st International Conference on Neural Information Processing Systems* (pp. 4768–4777). doi:10.5555/3295222.3295230

Marler, J. H., & Boudreau, J. W. (2017). An evidence-based review of HR analytics. *The International Journal of Human Resource Management*, 28(1), 3–26. doi:10.1080/09585192.2016.1244699

Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. *ACM Computing Surveys*, 54(6), Article 115. doi:10.1145/3457607

Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I. D., & Gebru, T. (2019). Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency* (pp. 220–229). doi:10.1145/3287560.3287596

Raji, I. D., Smart, A., White, R. N., Mitchell, M., Gebru, T., Hutchinson, B., Smith-Loud, J., Theron, D., & Barnes, P. (2020). Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency* (pp. 33–44). doi:10.1145/3351095.3372873

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). Why should I trust you? Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 1135–1144). doi:10.1145/2939672.2939778

Tambe, P., Cappelli, P., & Yakubovich, V. (2019). Artificial intelligence in human resources management: Challenges and a path forward. *California Management Review*, 61(4), 15–42. doi:10.1177/0008125619867910